Homework 6, due Oct 1 before class

1. A particle moves in the dipolar Earth magnetosphere. It is reflected at polar angle $\theta_r = \pi/4$. Find its pitch angle α at equator, Fig. 1.

   Hint: Particle moves along a given dipolar field line parametrized by $r = \sin^2\theta R$. Using expression for the strength of the dipolar magnetic field at given r and θ, use $r \to \sin^2\theta R$. This will give the magnetic field along a given field line. Use conservation of the adiabatic invariant to find $\alpha(\theta)$ (reflection point corresponds to $\alpha = \pi/2$).

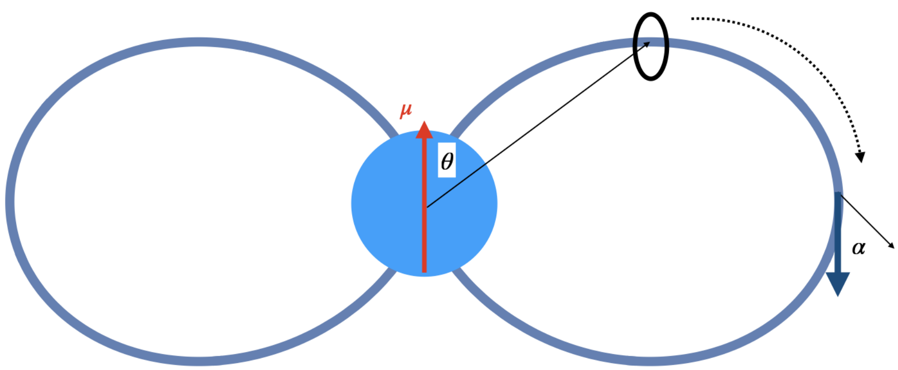

Fig. 1.— Bouncing particle in the dipolar magnetosphere. At the reflection point (a ring) pitch angle is $\alpha = \pi/2$.

2. Lorentz transformations for a particle in magnetic field.

   Consider a particle moving in magnetic field along helical trajectory with velocity β and pitch angle α, so that

$$
\begin{aligned}
&\mu = \cos\alpha \\
&\beta_\parallel = \beta\mu \\
&\beta_\perp = \beta\sqrt{1-\mu^2}
\end{aligned} \tag{1}
$$

   Make a boost with $\beta_\parallel$ to the frame K' where parallel velocity is zero, $\beta'_\parallel = 0$. (Prime denotes that frame K'). In the frame K' find velocity $\beta'_\perp$, momentum $p'_\perp$ and Lorentz factor $\gamma'_\perp$.

3. Lorentz transformations for fields.

   Magnetic dipole μ is aligned with z axis and is moving with velocity βc along x direction. For small $\beta \ll 1$ find electric field in lab frame.